2016.0 RANGE ROVER (LG), 415-01

INFORMATION AND ENTERTAINMENT SYSTEM

DIAGNOSIS AND TESTING

PRINCIPLE OF OPERATION

For a detailed description of the information and entertainment system, refer to the relevant description and operation sections in the workshop manual.

INSPECTION AND VERIFICATION

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle.

NOTES:

- If the control module or a component is suspect and the vehicle remains under manufacturer warranty, refer to the Warranty Policy and Procedures manual, or determine if any prior approval programme is in operation, prior to the installation of a new module/component.
- When performing voltage or resistance tests, always use a digital multimeter accurate to three decimal places, and with an up-to-date calibration certificate. When testing resistance always take the resistance of the digital multimeter leads into account.
- All diagnostic equipment should comply with local legislation.
- Relevant diagnostic equipment should be regularly checked and calibrated according to the manufacturer's instructions.
- The workshop should be equipped with a full range of general equipment which is to be kept in good order and available to all suitably trained staff.
- Diagnostic equipment must meet the JLR Minimum Standards for general equipment as outlined in TOPIx.
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.

1. Verify the customer concern

2. Visually inspect for obvious signs of damage and system integrity

Visual Inspection

MECHANICAL	ELECTRICAL
▪ Audio amplifier module ▪ Integrated audio module ▪ Digital radio control module	▪ Fuses ▪ Loose or corroded connector(s) ▪ Audio amplifier module

- Integrated control panel - Touch screen - Touch screen switchpacks - Navigation control module - Rear seat entertainment control module - Satellite radio control module - TV control module - Rear view camera - Compact disc player jammed, not loading - Scratched/dirty compact discs - Speakers	- Integrated audio module - Digital radio control module - Integrated control panel - Touch screen - Touch screen switchpacks - Navigation control module - Rear seat entertainment control module - Satellite radio control module - TV control module - Rear view camera - Speakers

3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step

4. If the cause is not visually evident, check for diagnostic trouble codes (DTCs) and refer to the relevant DTC index

GEN 2.1 AUDIO AMPLIFIER DIAGNOSTICS

The symptom chart below should be used when diagnosing audio amplifier module faults

REPORTED SYMPTOM	SYMPTOM DESCRIPTION	POTENTIAL CAUSES (FOR GUIDANCE ONLY)	RECOMMENDED ACTION
- Complete loss of audio from infotainment system	- No audio functions or no sound can be heard when audio function is selected	- Audio amplifier module not correctly installed - Circuit connectors to audio amplifier module disconnected or not correctly	- 1. Check TOPix for any audio system, audio amplifier module and MOST related SSM/TSBs and perform the specified rectifications as required. Retest to confirm if the issue has been resolved. If the fault is still present, go to next step - 2. Connect the manufacturer approved diagnostic system, check for DTCs and any
- Greyed out AV button -	- Audio functions are		

	No response to audio function commands	not available	secured ▪ MOST ring break ▪ Audio amplifier module fuse failure ▪ Data communication error between audio amplifier module and integrated audio module ▪ Power feed not present or power/ground circuit fault ▪ Audio amplifier module internal failure	recommended software updates. If DTCs are present or if software updates are required, perform the necessary remedial actions as specified by the SDD tool. After performing the necessary actions, confirm if the customer fault is still present. If no DTCs are logged and/or no software updates are required, go to the next step ▪ 3. Using the manufacturer approved diagnostic system, run the audio amplifier module On Demand Self-Test (ODST) routine and check for any DTCs that are logged. If DTCs are present, perform the necessary remedial actions as specified by the SDD tool. If the customer fault is still present after performing ODST routine and rectifying DTCs, go to the next step ▪ 4. Using the manufacturer approved diagnostic system, perform a vehicle reset by selecting the “Special Application – vehicle reset” routine. Check for DTCs and confirm if the fault is still present. If issue is still present, go to the next step ▪ 5. Gain access to the audio amplifier module and check the security and integrity of the cable connections to the audio amplifier module with reference to the circuit diagrams. Rectify as required ▪ 6. With the infotainment system switched on, disconnect the MOST connector from the audio amplifier module and check that a red light is present at the connector end. This will be steady for a second or two and then start flashing. If no light is present, there is a	

			problem in the MOST ring before the audio amplifier module. Other modules on the MOST ring must be investigated further ▪ 7. Perform an audio amplifier module hard reset by disconnecting the power cable to the audio amplifier module for at least 3 minutes. Reconnect the power cable and retest. If the fault is still present, go to the next step ▪ 8. Replace the audio amplifier module and configure as per the instructions outlined on the SDD tool. Confirm if the fault is still present. If issue is still present go to next step ▪ 9. If the fault is still present following replacement of the audio amplifier module, contact JLR Dealer Technical Support following the guidelines in the policy and procedures manual

▪ Loss of audio from one or more channels	▪ Partial loss of sound	▪ Audio amplifier module not correctly installed
		▪ Circuit connectors to audio amplifier module disconnected or not correctly secured
		▪ Incorrect EQ file installed on the audio amplifier module
		▪ Water ingress
		▪ Speaker fault
		▪ Data
			▪ 1. Verify the reported fault by performing the following checks:
			  ▪ Reset fade and balance settings to the centre of the vehicle
			  ▪ Confirm if the issue reported on the customer’s vehicle occurs just after engine start up or after the audio system has been running for 10 minutes or more
			  ▪ Confirm if the fault is present on all audio sources (CD, Radio, DAB Radio/SDARS, USB, Aux) – if the fault is not present on all sources, investigate affected audio source e.g. if issue is only seen on DAB Radio, check DAB module is functioning

▪ Audio output crackle or distortion	▪ No significant extraneous noise, but the audio reproduction is not as expected e.g. it is not clean sounding		

			communication error between audio amplifier module and integrated audio module ▪ Incorrect software installed on the audio amplifier module ▪ Intermittent ground circuit connection fault ▪ Audio amplifier module internal failure	correctly and the software is up to date ▪ Confirm if the issue is seen on stereo and surround sound settings (if applicable) ▪ 2. Check TOPix for any audio system, audio amplifier module and MOST related SSM/TSBs and perform the specified rectifications as required. Retest to confirm if the issue has been resolved. If the fault is still present, go to next step ▪ 3. Connect the manufacturer approved diagnostic system, check for DTCs and any recommended software updates. If DTCs are present or if software updates are required, perform the necessary remedial actions as specified by the SDD tool. After performing the necessary actions, confirm if the customer fault is still present. If no DTCs are logged and/or no software updates are required, go to the next step ▪ 4. Using the manufacturer approved diagnostic system, run the audio amplifier module On Demand Self-Test (ODST) routine and check for any DTCs that are logged. If DTCs are present, perform the necessary remedial actions as specified by the SDD tool. If the customer fault is still present after performing ODST routine and rectifying DTCs, go to the next step ▪ 5. Using the manufacturer approved diagnostic system, perform a vehicle reset by selecting the “Special Application – vehicle reset” routine. Check for DTCs and confirm if the fault is still	

			present. If issue is still present, go to the next step ▪ 6. Gain access to the audio amplifier module and check the security and integrity of the cable connections to the audio amplifier module with reference to the circuit diagrams. Rectify as required ▪ 7. With the infotainment system switched on, disconnect the MOST connector from the audio amplifier module and check that a red light is present at the connector end. This will be steady for a second or two and then start flashing. If no light is present, there is a problem in the MOST ring before the audio amplifier module. Other modules on the MOST ring must be investigated further ▪ 8. Perform an audio amplifier module hard reset by disconnecting the power cable to the audio amplifier module for at least 3 minutes. Reconnect the power cable and retest. If the fault is still present, go to the next step ▪ 9. Replace the audio amplifier module and configure as per the instructions outlined on the SDD tool. Confirm if the fault is still present. If issue is still present go to next step ▪ 10. If the fault is still present following replacement of the audio amplifier module, contact JLR Dealer Technical Support following the guidelines in the policy and procedures manual

MICROPHONE DIAGNOSTICS

The symptom chart below should be used when diagnosing microphone faults

REPORTED SYMPTOM	SYMPTOM DESCRIPTION	POTENTIAL CAUSES (FOR GUIDANCE ONLY)	RECOMMENDED ACTION

▪ Voice command is not working	▪ When pressing the voice command button on the steering wheel, the system is not responding	▪ Issue with steering wheel switches
		▪ No microphone(s) installed
		▪ Microphone not connected correctly
		▪ Faulty microphone
			▪ 1. Check and confirm customer's vehicle does have voice command capability installed on the vehicle and that the voice command function is enabled
			▪ 2. The microphone is shared between the Bluetooth telephone function and the voice command function. Check that the voice command function works by giving a voice command, for example "Telephone Help" or "Navigation Help". Check that Bluetooth telephone calls can be made. Ensure that the following criteria are satisfied before any telephone call is attempted:
			  ▪ The telephone handset and associated level of software is included on the JLR approved list
			  ▪ The telephone/device battery is fully charged and in a serviceable condition
			  ▪ There is a reliable telephone network reception signal of suitable strength
			  ▪ The telephone/device is placed within the vehicle cabin area
			  ▪ The telephone/device is connected to the vehicle via Bluetooth
			▪ **If the voice command function works and the phone call receiver can hear the caller adequately when calls are**

▪ Bluetooth telephone – 3rd party hears interference	▪ When customer uses the telephone via a Bluetooth connection, the call receiver hears interference over the caller's audio signal	▪ The phone/device is incompatible with JLR infotainment system ▪ Poor (sub-optimal) placement of the phone/device within the vehicle ▪ Poor mobile phone network reception ▪ High mobile phone network demand	
▪ Bluetooth telephone – 3rd party cannot hear caller at all	▪ When customer uses the telephone via a Bluetooth connection, the call		

		receiver cannot hear the caller	▪ Damaged microphone harness
			▪ Microphone unclipped from housing
			▪ No microphone(s) installed
			▪ Microphone not connected correctly
			▪ Faulty microphone
				made from the vehicle, this suggests that the fault is not with the microphone
				▪ 3. If the phone call was unsuccessful and the voice command is inoperative, first check if there are any SSMs/TSBs for voice command functions, microphones or Bluetooth and perform the specified rectifications as required. Retest to confirm if the issue has been resolved. If the fault is still present, go to next step
				▪ 4. Connect the manufacturer approved diagnostic system, check for related DTCs and any recommended software updates. If DTCs are present or if software updates are required, perform the necessary remedial actions as specified by the SDD tool. After performing the necessary actions, confirm if the customer fault is still present. If no DTCs are logged and/or no software updates are required, go to the next step
				▪ 5. Gain access to the microphone to check the following:
				  ▪ Check the microphone is fitted to driver's side of the vehicle and rectify as required
				  ▪ Check the harness assembly is not trapped and rectify as required
				  ▪ Check the harness assembly connections are fully inserted and secured to the microphone and rectify as required
				  ▪ Check for any harness damage and repair as required
				▪ 6. Check the microphone for the following and replace the component if any damage is evident:	

			- Check the microphone connector is free from damage and corrosion - Check for visual damage to the microphone e.g. wiring or casing - 7. If the fault is still present following replacement of the microphone, contact JLR Dealer Technical Support following the guidelines in the policy and procedures manual

WHITEFIRE WIRELESS HEADPHONES – FURTHER DIAGNOSTICS

For diagnosis of issues with wireless headphones operation, see symptom chart below:

NOTES:

- If a headphone concern is identified on a vehicle based in the United Kingdom market, please submit an Electronic Product Quality Report (ePQR) prior to carrying out diagnosis as JLR engineering would like to visit the vehicle to assist with diagnosis. If the customer is not prepared to wait for an extended period of time to allow for an engineering visit, then please continue with diagnosis and rectification
- There are up to three audio channels available for wireless headphones, depending on the vehicle specification. 1 - wireless headphones (dual view touch screen output only); 2 - wireless headphones (rear seat entertainment output only); 3 - wireless headphones (dual view touch screen and rear seat entertainment output)

SYMPTOM	POSSIBLE	ACTION

	CAUSES	
Wireless headphones status LED is permanently off	▪ Wireless headphones have not been switched on	▪ Press the power switch on the right side of the wireless headphones
	▪ No batteries are installed in the wireless headphones	▪ Remove battery cover on the left side of the wireless headphones and insert 2 x new AAA batteries
	▪ Batteries are inserted with incorrect polarity	▪ Check the batteries are installed as per the diagram in the battery compartment
	▪ Batteries are depleted	▪ Remove battery cover on the left side of the wireless headphones and replace batteries with 2 x new AAA batteries
	▪ Wireless headphones have timed out and automatically switched off	▪ Press the power switch on the right side of the wireless headphones
If the headphones status LED remains inactive following completion of all remedial actions detailed above then replace the headphones		
Wireless headphones status LED flashing (red or amber) and no audio from headphone. This indicates the wireless headphones are switched on ready to receive a signal from the vehicle overhead console emitter/ infotainment system	▪ Overhead console emitter is not active (an array of red LEDs can be seen when active)	▪ Ensure the vehicle infotainment system is on and the dual view and/or rear seat entertainment features are active ▪ Select the same audio source to cabin speakers to confirm audio output is operational ▪ If the vehicle overhead console emitter LEDs are not activated following completion of the remedial actions detailed above then continue with diagnosis of the audio system with reference to TOPIx and via Symptom Driven Diagnostics (SDD) which may be

		accessed using the manufacturer-approved diagnostic tool
	▪ Wireless headphones are not in direct line of sight with the overhead console emitter	▪ Ensure there is a direct line of sight between the overhead console emitter and the wireless headphones
	▪ Wireless headphones are not being used inside the vehicle	▪ The wireless headphones are designed to be used whilst seated in the vehicle only
If, following completion of all remedial actions detailed above, there is no audio output from the headphones when the overhead console emitter LEDs are active and there is a direct line of sight between the overhead console emitter and the wireless headphones, continue to next step		
Wireless headphones status LED constantly on (red or amber) and no audio output from wireless headphones. This indicates wireless headphones are switched on and receiving a signal from the entertainment system via the overhead console emitter	▪ Audio source selected is not playing (CD paused or radio station not tuned in)	▪ Ensure the active audio source is playing ▪ Select the same audio source to cabin speakers to confirm audio output is operational
	▪ Volume control is set to minimum	▪ Rotate the volume control located on the right side of the wireless headphones in a clockwise direction to increase volume
	▪ Wireless headphones are set to an inactive zone channel. (There are up to 3 channels depending on vehicle specification - see note below)	▪ Press the channel select switch on the right side of the wireless headphones. A beep will be heard from the wireless headphones when the channel is changed. Repeat if necessary

	▪ Wireless headphones audio is inhibited in dual view touch screen or rear seat entertainment control menu	▪ Enable the wireless headphones audio via the dual view touch screen or rear seat entertainment control menu
If there is no audio output from the headphones (or audio output from only one earpiece) following completion of all remedial actions detailed above then replace the headphones		

SPEAKER DIAGNOSTICS

For Speaker Diagnostics please refer to section 415-01 Diagnosis and Testing - Speakers in the workshop manual.

REFER to: Speakers (415-01 Information and Entertainment System, Diagnosis and Testing).

CAMERA DIAGNOSTICS

For Camera Diagnostics please refer to section 413-13 Diagnosis and Testing - Proximity Camera in the workshop manual.

REFER to: Proximity Camera (413-13 Parking Aid, Diagnosis and Testing).

DTC INDEX

For a list of diagnostic trouble codes that could be logged on this vehicle, please refer to Section 100-00.